A CATHOLIC INTRODUCTION TO ARTIFICIAL INTELLIGENCE · CLASS 7

AI in Industry, Healthcare & Finance

"AI is a witness to truth at best, never the source of truth."



This week in *Magnifica Humanitas*: the dignity of work at a time of digital transition.



TODAY

Agenda

1 Class 6 Assignment Review

2 Magnifica Humanitas

3 Class 7 Lecture

4 Class 7 Assignments



Opening Prayer

O Glorious St. Joseph, model of all those who are devoted to labor, obtain for me the grace to work conscientiously, putting the call of duty above my natural inclinations; to work with gratitude and joy, in a spirit of penance for the remission of my sins, considering it an honor to employ and develop by means of labor the gifts received from God; to work with order, peace, moderation, and patience, without ever shrinking from weariness and difficulties; to work above all with purity of intention and detachment from self, having always death before my eyes and the account that I must render of time lost, of talents wasted, of good omitted, of vain complacency in success, so fatal to the work of God. All for Jesus, all through Mary, all after thine example, O Patriarch, St. Joseph. Such shall be my motto in life and in death. Amen.



PART 1 OF 4

Class 6 Assignment Review

Truth, bias, ethics and morality



CLASS 6 IN REVIEW

Wisdom comes from persons, not machines

Truth

AI predicts patterns from its training data. It can sound convincing even when it is wrong.

Bias

Models can reflect the cultural biases in their data, and confidently present false or questionable content.

Persons

Made in the image of God, we have reason, conscience and moral agency. AI has none of these.

Your responsibility

Know when AI helps, when to question it, and when not to use it at all.



MODEL ANSWER · QUESTION 1

Long Deadlines and Lateness

THE QUESTION

The model learned that students with more days until the deadline are slightly more likely to be late. Why might that be? What does it tell us about procrastination and human behavior?

Students with more days until the deadline may be late more often because a distant due date feels less urgent, so they put off starting the assignment. As the deadline approaches, other assignments and activities compete for their attention, and the task may be forgotten or rushed at the last minute. The model reflects this pattern: `days_until_due` has a coefficient of −0.177, a slight effect, which means it is a real tendency but not a rule for every student. This shows that people often judge tasks by how urgent they feel rather than how important they are, which is the heart of procrastination.

WHAT MAKES IT STRONG

1. Answers the question in the first sentence
2. Cites the model's evidence and calls the effect "slight"
3. Ends by explaining human behavior

MODEL ANSWER · QUESTION 2

Which Error Is Worse?

THE QUESTION

Which type of error concerns you more: missing students who are actually late, or wrongly flagging students who are on time? What would the real-world consequences of each error be in a school setting?

Missing students who are actually late concerns me more than wrongly flagging students who are on time, because the purpose of this system is to find students who need help. A missed student receives no reminder or support, so they may fall behind or lose points to a late penalty that a simple check-in could have prevented. A wrongly flagged student receives an unnecessary check-in, which may be mildly embarrassing or make them feel doubted, but a short conversation usually clears it up. In this model, the imbalance is striking: at the default threshold, 13 of the 14 late students were missed, while only 1 on-time student was wrongly flagged.

WHAT MAKES IT STRONG

1. States the position first and restates the question
2. Gives a concrete consequence for each error
3. Uses the notebook's numbers: 13 missed, 1 flagged

MODEL ANSWER · QUESTION 3

Choosing a Threshold

THE QUESTION

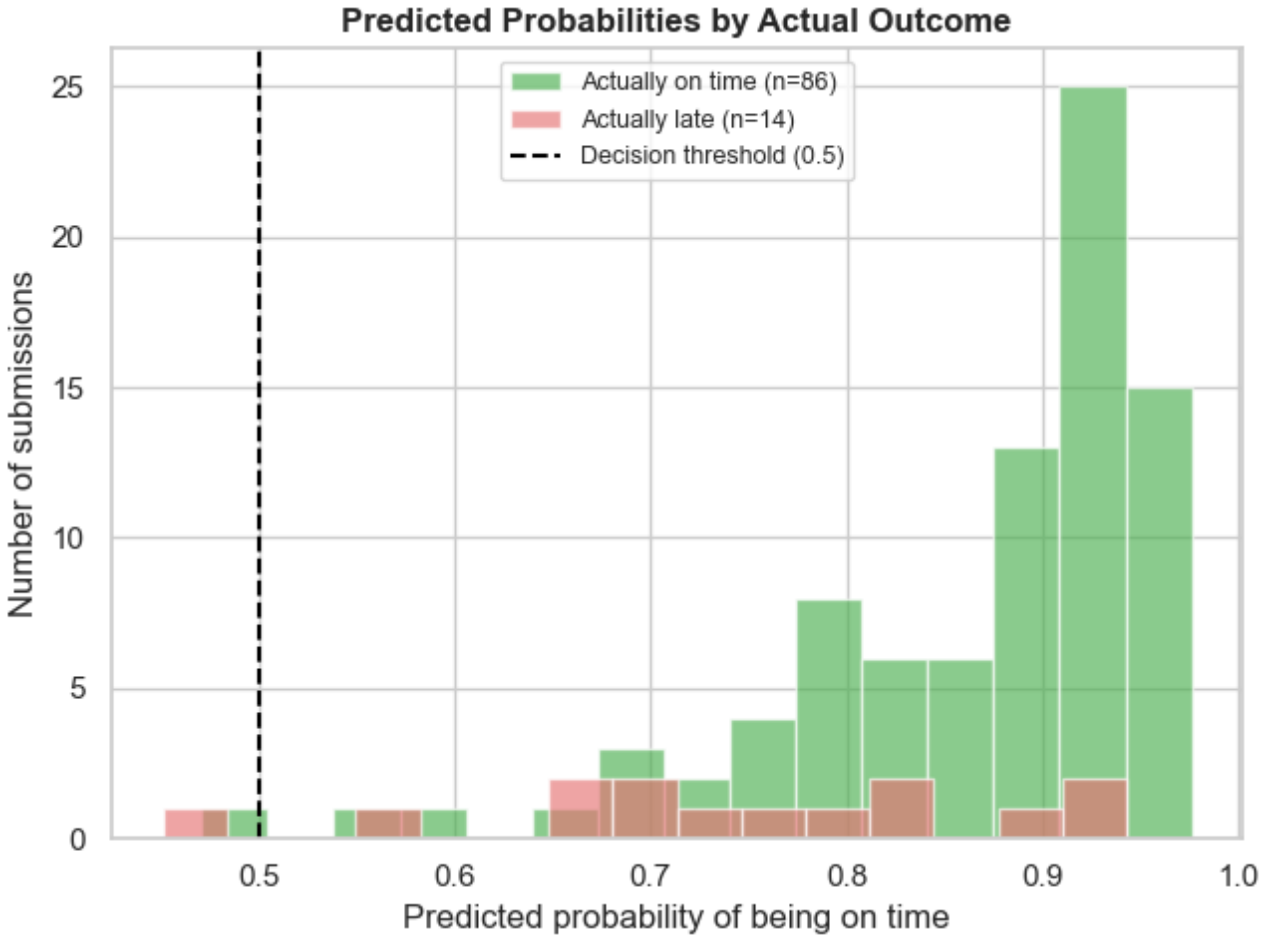
If you were a teacher deploying this system, what threshold would you set, and why? What would it mean to flag 40% of students as "at risk" in order to catch 80% of the genuinely late ones?

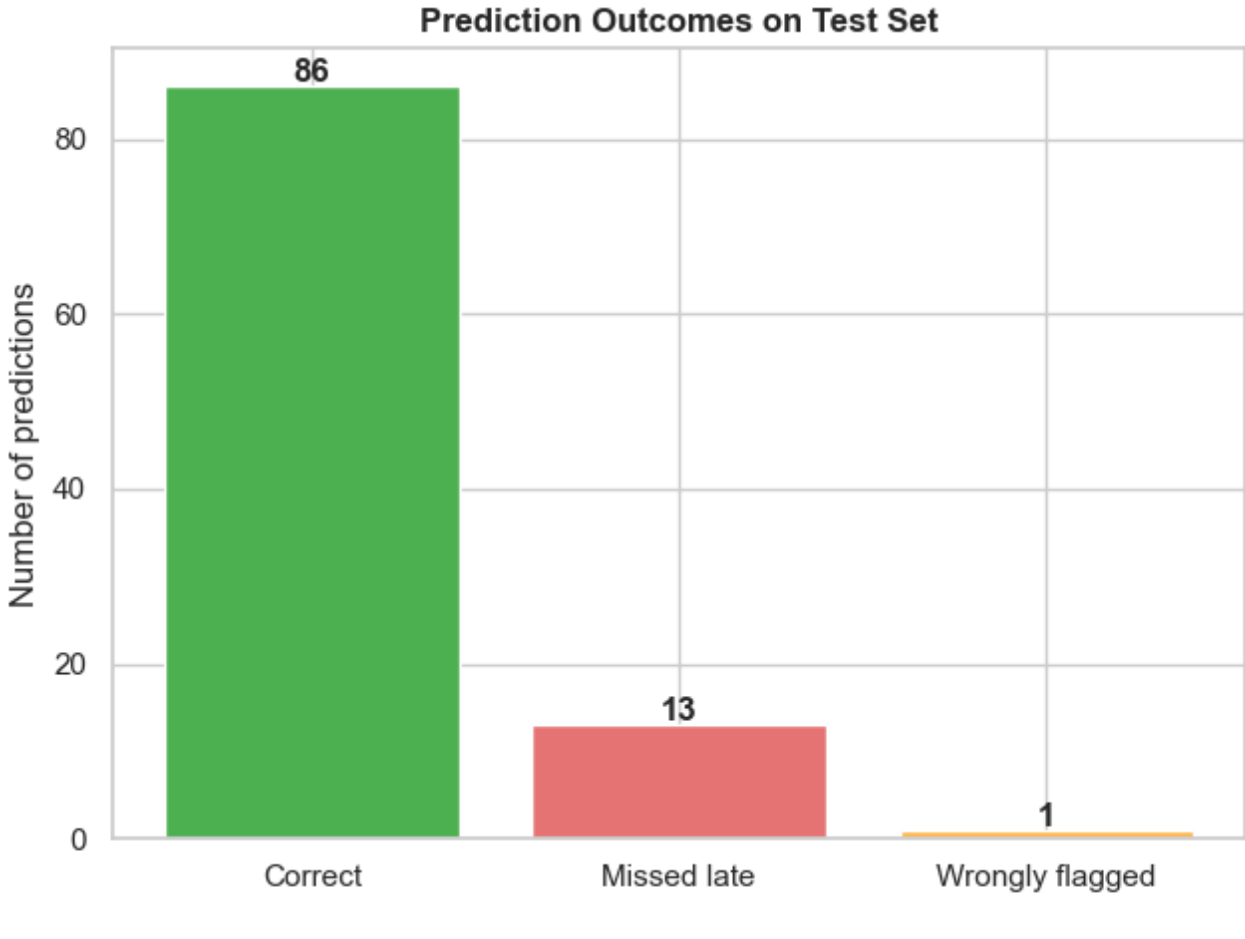
I would set the threshold at 0.70, because it catches far more late students than the default while keeping false alarms low. At 0.70, the model catches 6 of the 14 late students (43%) and wrongly flags only 5 of the 86 on-time students, and its accuracy (87%) is slightly better than always guessing "on time." Flagging 40% of students to catch 80% of the late ones, the 0.85 row, would mean 39 flags, 28 of them for students who were already on time. That workload would pull the teacher's time away from students who are truly struggling and could teach everyone to ignore the flags. Whichever threshold is chosen, each flag should lead to a teacher's review and a friendly check-in, never an automatic consequence.

WHAT MAKES IT STRONG

1. Names a specific threshold
2. Reads the table correctly: caught vs. flagged
3. Keeps a person in charge of every flag

Choosing a Threshold





LAB DEBRIEF

Reading the Threshold Table

Threshold	Late students caught (of 14)	On-time students flagged (of 86)	Total flagged (of 100)
0.50	1	1	2
0.70	**6**	**5**	**11**
0.80	8	20	28
0.85	**11**	**28**	**39**
0.90	12	45	57
0.95	14	76	90

Counts from the test set of 100 students. Highlighted rows: the two thresholds most worth comparing.

1

Read across one row

Each row is one choice of threshold. Its numbers belong together.

2

Higher threshold, more flags

The threshold is the chance of being on time, so raising it flags more students.

3

Caught is not the same as flagged

At 0.85, 39 students are flagged, but only 11 of them were late.

LAB DEBRIEF

What a Flag Means

1

The model predicts
Before the due date, it estimates who might turn work in late.

2

A teacher reviews
A person looks at the flag and the student's situation.

3

A friendly check-in
The student is offered a reminder or help.

A flag is never a grade, a punishment, or an accusation.

SAY THE ERRORS IN PLAIN WORDS

Missed a late student
The model said on time; the student was late. They never got the check-in.

Wrongly flagged an on-time student
The model said late; the student was on time. They got a check-in they didn't need.

LAB DEBRIEF

Writing a Strong Short Answer

1. Answer the question in your first sentence
2. Restate the question in your answer
3. Write 3–5 complete sentences
4. Cite one number from the notebook
5. Write formally and proofread

BEFORE

“Well, people procrastinate a lot, so it’s kind of like that.”

AFTER

“Students with more days until the deadline may be late more often because a distant due date feels less urgent. The model’s coefficient for days until due is −0.177, a slight effect, so this is a tendency, not a rule.”

Pulse check

- Do you understand what we just went over?
- A – totally confused
- B – somewhat confused
- C – I understand
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PART 2 OF 4

Magnifica Humanitas

The dignity of work at a time of digital transition

Pulse check

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PART 3 OF 4

Class 7 Lecture

AI in industry, healthcare and finance: separating reality from hype



LEARNING OBJECTIVES

By the end of class, you should be able to...

1

Identify several major industries currently using AI.

2

Explain how AI improves efficiency or decision-making in those industries.

3

Describe real-world risks and limitations of industrial AI systems.

4

Recognize how bias, automation and overreliance on AI can affect society.

5

Evaluate AI systems using ethical and faith-informed reasoning.

6

Research and communicate how AI is applied in a chosen field.





OVERVIEW

Where AI is already at work

Healthcare

Reading scans, scoring risk, drafting records

Finance

Fraud detection, lending, trading, insurance

Agriculture

Precision farming, crop health, weather

Manufacturing

Predictive maintenance, quality control

Transportation

Route optimization, vehicle safety

Customer service

Chatbots and virtual assistants

For each one, ask: what does it do well, what are its limits, and does it serve the person?





HEALTHCARE

AI reads the x-ray first

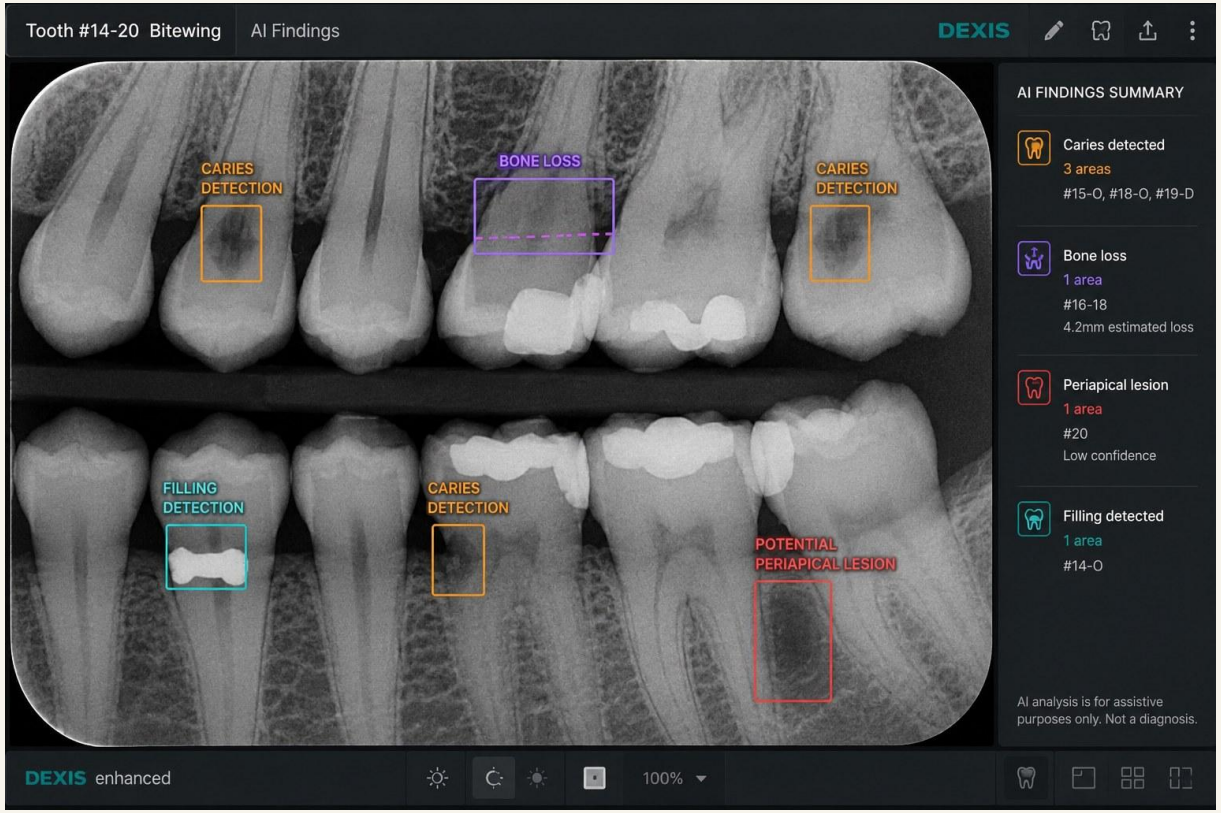


Illustration of an AI-annotated dental x-ray

At a routine checkup, each x-ray appears on a monitor already marked with color-coded labels.

How it works

The AI compares the image against a baseline of thousands of similar x-rays and flags possible trouble spots.

> *The AI highlights. The dentist diagnoses.*





HEALTHCARE

From 45 minutes to seconds

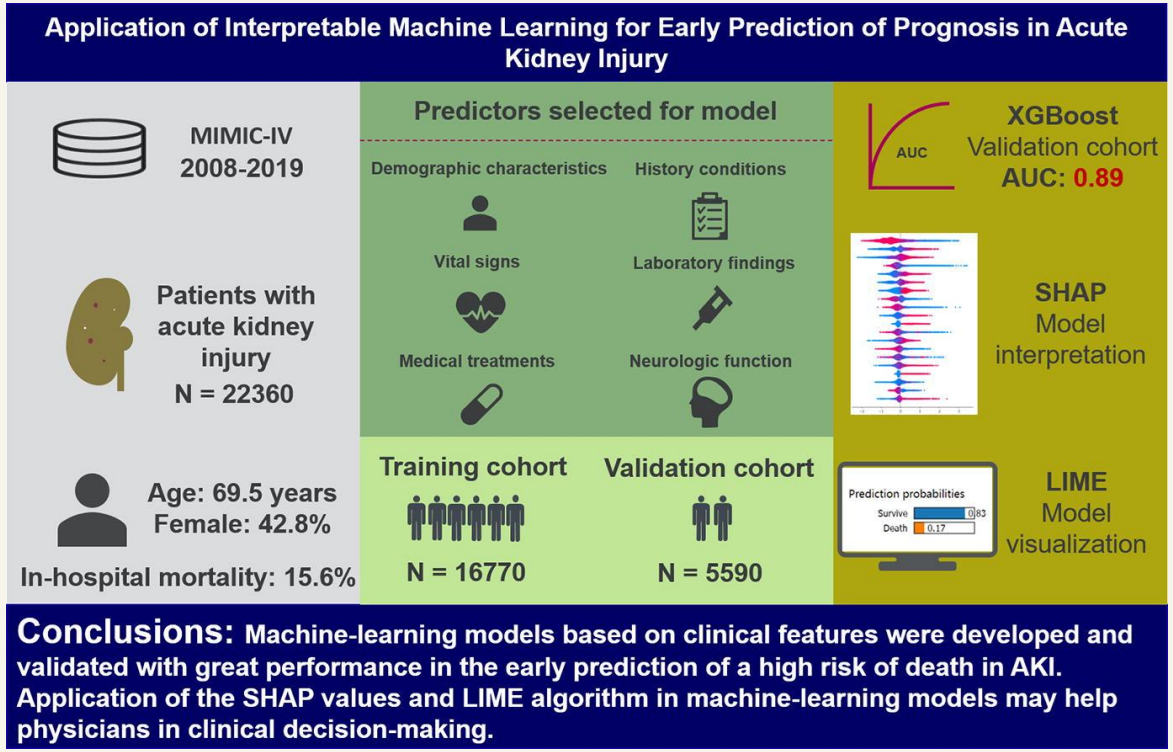
Total kidney volume

45 min

by hand, per patient, reviewing dozens of kidney images one after another

seconds

with AI automating the same measurement



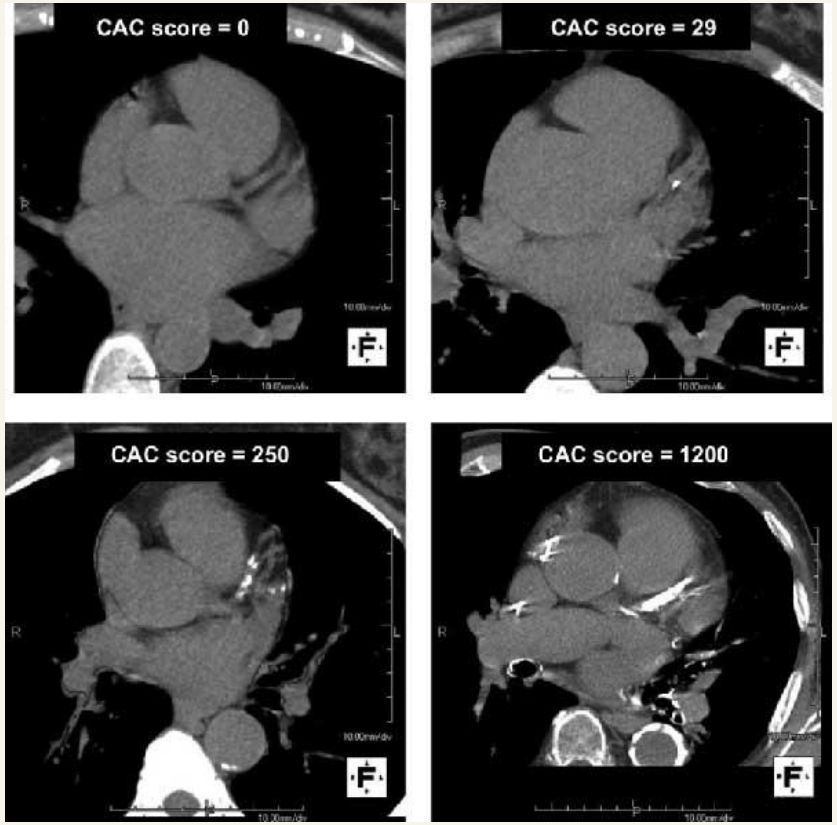
A study predicting risk in acute kidney injury: AUC 0.89, plus tools (SHAP, LIME) that show doctors why the model decided.





HEALTHCARE

Scoring heart risk from a CT scan



1. **Scan**
   A CT scan images the heart's arteries.

2. **Compare**
   An AI model trained on thousands of scored images calculates a coronary calcium score (CAC).

3. **Predict**
   The score estimates the risk of a heart attack or stroke over the next 5 to 10 years.

4. **Treat**
   The doctor uses the score to design a plan that lowers that risk.





HEALTHCARE

Paperwork and support

Clinical notes and forms

As a professional types, the software reads the patient's history, proposes how to finish the entry and pre-fills other fields. The professional must review and approve everything before saving.

AI as an informal counselor

Chatbots are widely used for support by stressed young people and lonely older people. Israel's Sheba Medical Center built LIV, a set of mental-health AI tools; an AI avatar has improved outcomes for people with PTSD.

Helpful in high-stress moments, but a simulated friend is not a real one.





HEALTHCARE

Healthcare professionals are still needed

Clinical context

for the algorithms that train AI

Accurate information

relevant data for AI to analyze

Translation

of AI findings into something meaningful for patients

AI can make care faster. Only a person can care for a patient.





FINANCE

How AI is used in finance

Fraud detection

Learns a customer's normal activity and flags unusual transactions.

Credit & lending

Estimates whether a borrower will repay, from history, income and debt.

Investment

Analyzes markets and news; automated trading reacts in fractions of a second.

Customer service

Chatbots answer questions any time of day.

Insurance

Assesses risk, sets premiums, processes and checks claims.

Finance runs on data, and AI is built to find patterns in data.





FINANCE · CASE STUDY

Catching credit card fraud

	Decision tree (baseline)	Decision tree + class weight	Decision tree (tuned)	Random forest (baseline)	Random forest (tuned)
Accuracy	0.958	0.962	0.967	0.979	0.984
Recall	**0.659**	**0.660**	**0.664**	**0.657**	**0.660**
Precision	0.522	0.565	0.621	0.836	0.975
F1	0.583	0.609	0.642	0.736	0.788

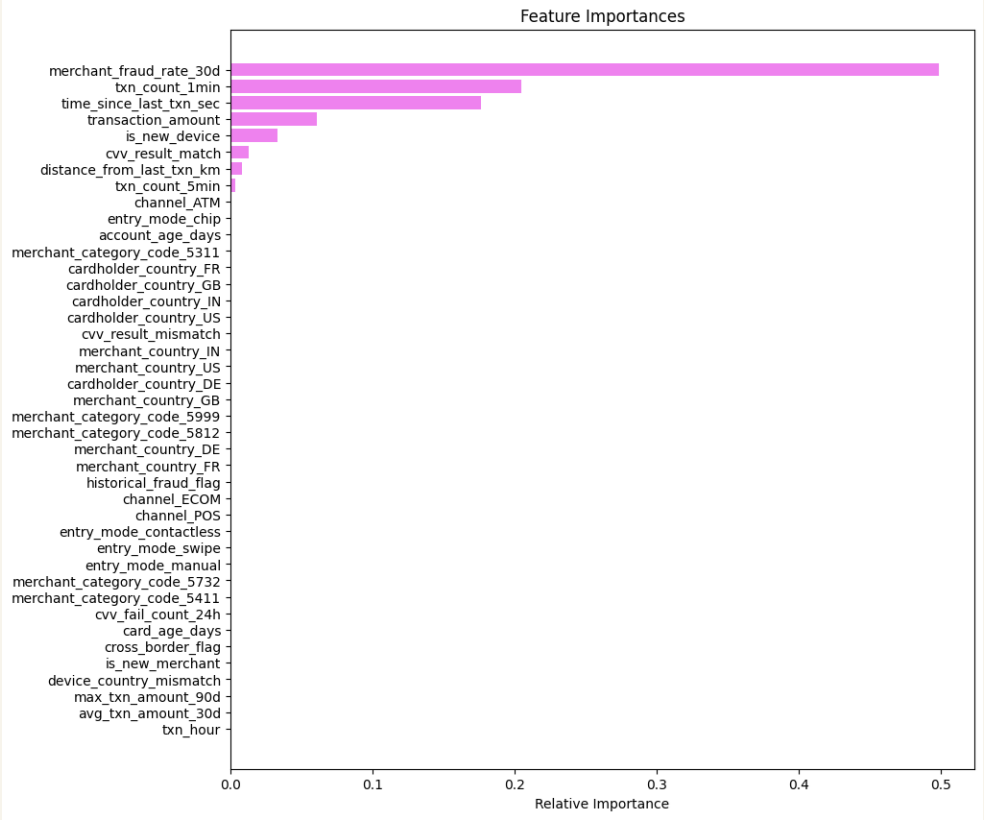
Every model is over 95% accurate, **yet each catches only about two out of three frauds.** Accuracy hides what a model misses.





FINANCE · CASE STUDY

What the model relies on



Reading the chart

- One feature, the merchant's fraud rate over 30 days, carries about half the model's importance.
- Next come how many transactions happened in the last minute and the time since the last one.
- Most of the other features barely matter.

Discuss: your card is declined because of where you shopped, not anything you did. Is that fair? Who can you appeal to?





FINANCE

Catholic social teaching and AI in finance

Human dignity

Persons are ends, not data points. Decisions about housing, credit or jobs need human oversight and a way to appeal.

Preferential option for the poor

AI should reduce economic exclusion, not entrench it.

Solidarity & the common good

Put stability, fairness and public trust ahead of short-term profit.

Stewardship & prudence

Manage data and computing power responsibly: safe, explainable, auditable.

In practice: human review for critical decisions · impact assessments · explanations and appeals · bias audits and diverse data · privacy and real consent · rules against systemic risk





AGRICULTURE

Farming with better eyes

Precision farming

Sensors and satellites show which parts of a field need more water or less fertilizer, reducing waste.

Crop monitoring

Drones and cameras spot disease, pests and nutrient problems earlier than the human eye.

Weather & yield prediction

Forecasts, climate history and soil data warn of drought, flooding, frost or heat stress.

Machinery & robotics

Self-driving tractors, fruit-picking and weeding robots, robotic milking and livestock monitoring.

Watch: AI weeding a field with lasers instead of herbicides

Watch: self-driving tractors



AGRICULTURE AS VOCATION

More than an industry

Agriculture is not merely an industry; it is a vocation connected to food, land, family, community and creation.

AI should respect farmers, protect workers, care for animals, preserve the environment and promote food security.

The real question: not only whether AI makes farming more productive, but whether it helps agriculture serve life, justice and the common good.



MANUFACTURING & TRANSPORTATION

AI in the physical world

The smart factory

Predictive maintenance

Sensors track temperature, sound and vibration so AI can warn before a machine fails, protecting workers from sudden breakdowns.

Quality control

Computer vision inspects items on fast assembly lines, from microchip flaws to sealed jars of baby food.

Moving the world

Route optimization

Traffic, weather and history chart the fastest, most fuel-efficient paths for trucks, ships and planes.

Safety & autonomy

Emergency braking and lane-keeping react faster than a foot can hit the brake. Fully self-driving cars are still being refined.





A CATHOLIC PERSPECTIVE

Dignity, creation and the common good

The dignity of work

Work is for the person, not the person for work. Let AI take dull, dirty and dangerous tasks, and retrain workers it displaces.

Care for creation

Laudato Si' calls us to protect our common home. Saving fuel and cutting waste make AI a tool of stewardship.

The common good

Measure these systems by whether they make roads safer, work more dignified and resources more sustainable.

Automation must never discard workers or treat them as mere cogs in a machine.



REFLECTION

Take these with you

Would you personally trust AI to help make decisions in healthcare or finance? Why or why not?

Pope Leo says a civilization is measured by “the care it is able to offer.” Where could AI help us care for one another better, and where could it get in the way?



VOCABULARY · 1 OF 2

Key terms

Predictive Analytics

Using AI to analyze data and predict future outcomes.

Fraud Detection

AI that identifies unusual or suspicious activity to prevent theft or fraud.

Medical Diagnosis

Using AI to help doctors identify diseases or health issues from medical data.

Automation

AI performing tasks without human intervention, often to save time or reduce errors.

Computer Vision

AI that lets machines “see” and interpret images or video.





VOCABULARY · 2 OF 2

Key terms

Recommendation System

AI that suggests products, services or actions based on data patterns.

Decision Support System

AI that assists humans in making better decisions with analyzed data and suggestions.

Algorithmic Bias

When AI unfairly favors or discriminates against certain groups because of biased training data.

Efficiency

How AI helps complete tasks faster, more accurately or with less waste.

Human Oversight

People supervising and approving AI decisions to ensure accuracy and ethics.



Pulse check

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PART 4 OF 4

Class 7 assignments

Weekly Quiz

Generate your Gemini Notebook interactive report/learning guide for the week. Use it to help you review and understand the material and complete the assigned quiz.

Lab: MoodBeats

Can a neural network hear a song's mood? Open class_07_lab_moodbeats.ipynb, train a neural network, compare it with logistic regression, and answer the reflection questions.



Prayer to St. Michael the Archangel

St. Michael the Archangel, defend us in battle. Be our protection against the wickedness and snares of the Devil. May God rebuke him, we humbly pray, and do thou, O Prince of the heavenly hosts, by the power of God, cast into hell Satan, and all the evil spirits, who prowl about the world seeking the ruin of souls. Amen.



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